Financial RAG — Noun-Phrase Collision Elimination + Chain-of-Verification

Full System Architecture: Offline Indexing Plane + Online Query Plane

OFFLINE PLANE — runs on corpus change
Path A inputs: PDF / HTML / DOCX / TXT / PPTX
Scanned — no text layer?
yes
no
OCR + table structure recovery PaddleOCR PP-Structure
Path B inputs: XBRL / CSV / XLSX / DB export
Schema Mapper maps columns/tags to canonical fields
Layout- & Table-Aware Parser format-driven, content-agnostic
Row-to-Proposition Templating string templating, no LLM
Proposition Segmenter **CALL SITE 1 — LLM** DenseX-style atomic segmentation structured output, may not alter digits
Numeric / Unit / Currency / Period Extractor **CALL SITE 2 — bounded LLM** tier1/2 regex first; LLM resolves qualifiers only — digits copied verbatim by code
Metadata Tagger from filing headers / XBRL tags
Three-index invariant: Dense, Sparse, Metadata Store index identical proposition_id sets
Index Writers deterministic + embedding model
Placeholder Registry Builder deterministic + embedding similarity clustering
Dense Index
Sparse Index — BM25
Metadata Store — Postgres/JSONB
Placeholder Registry
serves
serves
serves
serves
serves
C0 — NAIVE BASELINE (evaluation reference point only)
Fixed-size chunking (2000 chars, per page) · dense-only retrieval · no rerank · no Query Splitter · no Metadata Filter · no CoVe · no numeric-verbatim guard · separate index (financial_rag_c0) *Outside the NPCE x CoVe factorial — not part of the pipeline above.*
LEGEND
Deterministic code no model involved
LLM call — generative numbered = one of the six permitted call sites
Embedding / cross-encoder — non-generative neural model
Mixed: deterministic code + bounded/partial model use
Index / data store
Architectural note (on/off behaviour)
ONLINE PLANE — runs per question
User query
Query Analyzer noun-phrase extraction + collision detection (lexicon-only)
Query Splitter **CALL SITE 3 — LLM (slot parsing)** + DET (Cartesian product = the split itself)
Query-side Placeholder Remapper dictionary substitution
NPCE off (C1 / C2): exactly one unsplit sub-query — passthrough, not a code branch
Fan out: one branch per sub-query s_i
Metadata Filter SQL over Postgres
Hybrid Retriever BM25 (det) + Dense (embedding)
Rank Fusion + time-aware weighting
Cross-encoder Reranker non-generative model
FFF Rewriter **CALL SITE 4 — LLM** + DET numeric-verbatim guard (rejects any unsourced digit)
Fact-set / Query Pairer keyed binding only
Generator **CALL SITE 5 — LLM** answers from paired FFFs only
CoVe off (C1 / C3): draft D passes straight to the Abstention Gate
Draft Assembler templating; introduces no new facts → draft D
Chain-of-Verification — CALL SITE 6
Extract atomic claims from draft D **6a — LLM**
Template verification questions DET — deterministic construction
Execute verification independently **6b — LLM** draft D structurally absent from this call's context — not a prompt instruction
Claim type?
numeric
prose
Deterministic numeric comparison DET — value equality **never an LLM judgment**
Prose comparison **6c — LLM**
Revised / flagged claim set
Abstention Gate rules over verification results
Placeholder Reverse-Mapper restores real entity names
Citation Binder looks up proposition_id metadata **an LLM never writes a citation**
Answer + verification trace + citations (or abstention)